

HVE Core Challenge

From product intent to reviewed code

What HVE is

Hypervelocity Engineering is a structured engineering framework for GitHub Copilot.

It turns agent work into a repeatable process with explicit roles, standards, human gates, and durable evidence.

Four component types

Component	Job
Agents	Own roles and multi-step protocols
Prompts	Start repeatable workflows
Instructions	Apply standards by file pattern
Tracking artifacts	Carry state between phases

HVE uses files as interfaces

Research artifact

- Plan + critique
- Changes log
- Review findings and decision

Each handoff is inspectable, diffable, and resumable.

Research before code

Research answers:

- Which files own the behavior?
- What constraints already exist?
- Which alternatives were rejected?
- What check could disprove the approach?

Output: `.copilot-tracking/research/`

Plan before implementation

Plan maps criteria to files, checks, dependencies, and human gates.

Output:

```
.copilot-tracking/plans/
```

Implement and Review

Implement follows the approved plan and writes a changes log.

Review checks request fulfillment, code quality, tests, and instruction compliance.

The human still decides: approve, request changes, or pause.

Direct and combined RPI

Mode	Entry points	Use it when
Direct prompts	<code>/rpi-research</code> , <code>/rpi-plan</code> , <code>/rpi-implement</code> , <code>/rpi-review</code>	Learning, demos, explicit approvals
Combined	<code>/rpi task="..."</code>	The team understands and trusts the handoffs

Combined RPI uses Follow-up after Review to route defects or missing evidence.

HVE customization is additive

```
.github/copilot-instructions.md  
.github/instructions/*.instructions.md  
.github/agents/*.agent.md
```

Project context reaches all agents. Coding practices activate by `applyTo` glob.
Custom agents add repository-owned roles and output contracts.

Activation needs evidence

An instruction file existing is not proof.

Show:

1. a matching path;
2. the rule that activated;
3. the plan, code, or review decision it changed.

HVE starts before engineering

Business idea

- BRD Builder / Product Manager Advisor / PRD Builder
- Agile Coach
- RPI
- reviewed code

Product agents test value and assumptions. Agile Coach turns intent into an outcome-oriented story and acceptance criteria.

The challenge ramp

1. Run direct Research and Plan prompts.
2. Add repository context, coding practices, and a custom reviewer.
3. Use Product Manager Advisor and Agile Coach.
4. Update the plan, then run Implement and Review.
5. Compare the result with combined `/rpi`.

The implementation surface

Deliver `POST /api/decisions` from a bounded product intent and API contract.

The endpoint is small on purpose. Scope control is part of the score.

The contract

Input	Rule
title	Required non-empty string; duplicate check ignores case
context	Required non-empty string
decision	Required non-empty string
status	proposed, accepted, or superseded

Generate an ID and UTC timestamp. Keep state in memory.

Package tested source in a non-root runtime image with production dependencies and a health check.

Fast checks that find weak work

1. Send a required field as whitespace or a number.
2. Create the same trimmed title with different case.
3. Send malformed JSON.
4. Inspect the exact error envelope.
5. Check which stage supplies source to the runtime image.

These checks test the boundary, not the happy path.

Completion evidence

Area	Weight
HVE RPI artifact chain	40%
HVE customization and activation	25%
Product and Agile agent handoff	20%
Feature and container validation	15%

A manual implementation does not complete an HVE challenge.